

R&D Document: Setting up Point-to-Site (P2S) VPN using Azure Portal

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1.Introduction

With the growing use of cloud platforms like Microsoft Azure, secure remote access to cloud resources has become essential. Many organizations need a way for employees, developers, or students to connect to their private networks in the cloud from anywhere, especially when working remotely.

A Point-to-Site (P2S) VPN is a type of VPN connection that allows individual users to securely connect to an Azure Virtual Network (VNet) from their personal devices using the internet. It is simple to set up, cost-effective, and does not require any physical VPN hardware.

2.Objective

This document explores and demonstrates how to set up a Point-to-Site (P2S) VPN connection in Microsoft Azure using the Azure Portal. It is intended for securely connecting individual client systems (e.g., developer laptops) to a virtual network (VNet) hosted in Azure without the need for an on-premises VPN device.

3.What is Point-to-Site (P2S) VPN?

A Point-to-Site VPN is a secure connection between a client computer and an Azure virtual network. It is used when:

- You want to securely access Azure VMs or resources.
- You do not have an on-prem VPN device.
- You want individual client machines to connect remotely to Azure.

4. Components Required

Component	Description
Virtual Network (VNet)	The private network in Azure that we want to connect to
Gateway Subnet	Special subnet in VNet that holds the VPN Gateway
VPN Gateway	Azure-managed resource that allows VPN connections
Public IP	Required for gateway endpoint

Root & Client Certificates	Used to authenticate clients (Windows OS uses these)
VPN Client Configuration	To be downloaded and installed on local machine for connection

5.Step-by-Step Setup of P2S VPN on Azure Portal

Step 1: Create a Virtual Network (VNet)

1. Go to Azure Portal → Create a Resource → Networking → Virtual Network
2. Set:
 - Name: MyVNet
 - Address space: 10.1.0.0/16
 - Subnet name: Default
 - Subnet address range: 10.1.0.0/24
 - Region: Same region you'll create VPN Gateway in
3. Click Review + Create → Create

Step 2: Create a Gateway Subnet

1. Navigate to your created VNet → Subnets → + Gateway Subnet
2. Use address range: 10.1.255.0/27
3. Click Save

Step 3: Create a Public IP Address

1. Go to Create a Resource → Networking → Public IP Address
2. Set:
 - Name: MyVPNPublicIP
 - SKU: Standard
 - Assignment: Static
3. Click Create

Step 4: Create the VPN Gateway

1. Go to Create a Resource → Networking → Virtual Network Gateway
2. Set:
 - Name: MyVPNGateway
 - Region: Same as VNet
 - Gateway type: VPN
 - VPN type: Route-based
 - SKU: VpnGw1 (or Basic if minimizing cost)
 - Virtual Network: MyVNet
 - Public IP: MyVPNPublicIP
3. Click Review + Create → Create

Step 5: Generate Root and Client Certificates (on Windows)

1. Open PowerShell (Run as Administrator)

To Create a Root Certificate:

```
$cert = New-SelfSignedCertificate -Type Custom -KeySpec Signature `
-Subject "CN=MyRootCert" -KeyExportPolicy Exportable `
-HashAlgorithm sha256 -KeyLength 2048 `
-CertStoreLocation "Cert:\CurrentUser\My" `
-KeyUsageProperty Sign -KeyUsage CertSign
```

- Export Root Certificate (Base64):

```
$cert | Export-Certificate -FilePath "C:\Users\<username>\Desktop\MyRootCert.cer"
```

- To Create a Client Certificate:

```
New-SelfSignedCertificate -Type Custom -DnsName MyClientCert `
-KeySpec Signature -Subject "CN=MyClientCert" `
-Signer $cert -CertStoreLocation "Cert:\CurrentUser\My"
```

Step 6: Upload Root Certificate to Azure

1. Go to your VPN Gateway → Point-to-site configuration

2. Click Configure now
3. Set:
 - Address pool: 172.16.0.0/24
 - Tunnel type: IKEv2 and OpenVPN (SSL)
 - Authentication type: Azure certificate
 - Upload the base64-encoded public cert data (open .cer in Notepad and copy contents)
4. Click Save

Step 7: Download VPN Client Package

1. After saving P2S config, click Download VPN client
2. Extract the ZIP file

Step 8: Install & Connect VPN Client

1. Run the appropriate .exe file from the extracted VPN client package (based on your OS)
2. Go to Windows Settings → Network & Internet → VPN
3. You will see your VPN profile (e.g., MyVPNGateway)
4. Click Connect

6.Verification

1. After connecting, run ipconfig on command prompt → You should see IP from 172.16.0.x
2. Try pinging your Azure VM's private IP → If it replies, VPN is working

7. Testing

- Deploy a test VM in same VNet (MyVNet)
- Assign NSG rules to allow RDP (port 3389)
- Try connecting via Remote Desktop using the private IP

8. Use Cases of P2S VPN

Use Case	Description
Secure Developer Access	Developers can securely access test VMs in Azure
Remote Debugging	Connect to internal services without exposing them publicly
Training & Labs	Students/employees can connect to lab VMs from home

9.Limitations / Considerations

- Not suitable for large-scale connections (use S2S for that)
- Limited number of concurrent connections (based on SKU)
- Requires certificate management

10.Conclusion

Setting up a Point-to-Site VPN in Azure is a beginner-friendly and secure way to allow developers or admins to access cloud resources. As an intern, I was able to explore and implement this using certificates and verify the connection from my local system to the Azure VNet.